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Research Article



Generative AI (GAI) adoption in hotels and resorts: understanding competitive advantage using longitudinal & multi-group study

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ABSTRACT

This study examines the use of Generative AI (GAI) in the hospitality industry, particularly in hotels and resorts. It investigates the implications of GAI towards the competitive advantage, firm performance, and organisation's performance. Employing the Technological-Organizational-Environmental (TOE) framework, this study further explains how narcissism moderates the relationship between the factors that assist in adopting GAI and GAI's moderating effect on the overall competitive positioning and organisational outcomes. The study uses longitudinal analysis to shed some light about the influences related to GAI adoption, including the level of technological readiness in the organisation, as well as certain individual traits. The managers and owners will leverage these insights, addressing how GAI enhances operational efficiency, customer experience, and competitiveness in the hospitality industry context. This study combines the Technology-Organization-Environment (TOE) framework with the personality trait of narcissism to analyse the factors driving Generative AI adoption in the hospitality sector on different levels. Using a longitudinal approach, it provides new understanding of the impact of leadership characteristics on strategic technologies and management decisions, underscoring the value of leveraging GAI for sustained competitive edge and enhanced organisational efficiency.

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GAI; hotels & resorts; firm performance; competitive advantage

Statement of novelty

This study offers a novel contribution by being one of the first to empirically examine the adoption of Generative AI (GAI) within the hospitality industry through a longitudinal lens, integrating the Technology-Organization-Environment (T-O-E) framework alongside leadership traits such as narcissism. While most prior research has focused on static models or single-point analyses of AI adoption, this study uniquely captures how organisational readiness, technological capability, and environmental pressures evolve over time and influence competitive advantage and firm performance. By including psychological dimensions of leadership and using data from two different timeframes, the research provides a comprehensive and dynamic understanding of GAI adoption, offering both theoretical enrichment and actionable insights for industry stakeholders navigating digital transformation.

1. Introduction

Generative AI (GAI) is paramount to hotels and resorts, as it can enhance the guest experience and operational performance (Christensen et al., 2024; Dogru et al., 2025). Generative AI technologies

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allow the hospitality industry to improve customer interactions, including reservations and personalised out-of-room services (Ling et al., 2023; Wang, 2025). Further, with the help of generative AI, property managers can arrange appropriate resources and offer the right services to guests through analytics, creating operational and cost efficiency (Dogru et al., 2025; Li et al., 2024). In addition, it has been found that AI can make work easier for employees working in the hotel industry and reduce their mundane, repetitive work, in turn allowing them to give more attention to the guests, which will ultimately improve customer loyalty and induce positive word of mouth marketing (Singh & Singh, 2024).

As the emergence of technology is dictating consumer habits and the standards set within the industry, it is essential to consider what the rationale is behind the hospitality businesses' usage of generative AI in order to remain a competition (Dwivedi et al., 2024; Wong et al., 2023b). Investigating the factors influencing risk decision-making regarding generative AI adoption implementation, such as organisational readiness, technological capabilities, and environmental pressures, would help enhance the understanding of the risks and rewards of integrating AI in the hospitality sector (Dwivedi et al., 2024; Majid et al., 2024). Understanding this information will support the effective formulation of strategies, allocation of resources, and innovations, which will be crucial in pursuing long-term growth and success in a dynamic digital landscape (Hsu et al., 2024).

While many shareholders seem enthusiastic about using generative AI technologies, very few studies have directly addressed the factors upon which the hotels and resorts intend to embrace such key innovations (Law et al., 2023). What I see in most studies today are explorations into advanced AI technologies and their broader popularity as opposed to an increasing body of work on generative AI. Few studies investigate the role of organisational and technological contexts and environmental factors in developing the intention of hospitality stakeholders to invest in generative AI (Prasad Agrawal, 2024; Ivanov et al., 2024). On top of that, there seems to be little research that has sought to assess the implications of generative AI adoption, especially in terms of firm productivity, such as revenues, cost, and customer satisfaction, in the context of the hospitality industry.

The Technology-Organization-Environment (TOE) framework demonstrates all the elements of attention when trying to understand the degree of acceptance of the hotels or resorts towards generative AI technologies (Wang et al., 2016). Many studies, including the TOE framework, anomalously state that some factors impede the organisation's will to adopt change, such as generative AI adoption (Dwivedi et al., 2024). In the hospitality sector, the adoption of Generative AI (GAI) stems from the interrelation of technology, organisation, and an individual's level of involvement. Grasping these factors is vital in revealing the response of hotels and resorts towards digital transformation. This work takes a systematic approach to identify the guiding factors of GAI adoption, facilitating further exploration of the strategy, operation, and behaviour which motivate successful implementation.

This study leverages several pivotal constructs to understand the Generative AI adoption within the hospitality industry. It describes that organisational readiness is the level of a hotel or a resort's possession of the requisite resources, culture, and support frameworks for the efficient implementation and adoption of generative AI technologies (AL-Khatib, 2024; Darwish et al., 2025). Technological capability explains the propensity of an organisation to obtain, integrate, and effectively leverage sophisticated innovations like generative AI in improving operational processes and service delivery (Singh et al., 2024). Firm performance illustrates the holistic refractive decline in the efficacy of a hotel or resort towards key operational metrics like increased revenue, more efficient cost structure, heightened customer satisfaction, and favourable market positioning (Chatterjee et al., 2021a; Li et al., 2024). Narcissism defines a concept that is best known as a self-centred, self-admiring, and conceited personality trait that influences decision-making and risk-taking within organisations (Tandon et al., 2023).

Even though several personality factors affect the adoption of technology, this research deals with narcissism because it relates to leadership and risk-taking behaviour, as well as innovation adoption in organisations. Narcissistic people have an inflated self-image, a strong desire to be

recognised, and a tendency to make overly bold decisions (Campbell & Foster, 2011; Emmons, 1987). Such traits can have a pronounced impact on a person's response to disruptive innovations like Generative AI (GAI). Unlike openness to experience, which entails a willingness to learn and accept new ideas (McCrae & Costa, 1997), narcissism is an actively driving force of behaviour through a desire to reach goals, gain, and control especially in a leader's context (Rosenthal & Pittinsky, 2006). Often narcissistic managers are overconfident in making organisational decisions, actively seeking out new technologies, and striving to compete on differentiation—factors that are directly aligned with GAI's promise of strategic transformation (Chatterjee & Hambrick, 2007). In addition, higher innovation and digital leadership transformation through narcissism have been reported, especially in the context of high energy and highly visible domains such as hospitality (Engelen et al., 2016; Zhang et al., 2015). Thus, in the context of the Technology-Organization-Environment (TOE) framework, this research treats narcissism as a moderating variable in an attempt to reveal how personality traits at the individual level interrelate with organisational and technological factors to produce certain outcomes with respect to adoption—a gap that continues to persist within the literature on AI adoption.

Individual differences, especially personality traits, have been known to influence decision-making, particularly regarding new technologies such as GAI. For instance, an individual with a high level of narcissism is likely to show higher self-confidence, a higher tendency to take risks, and a greater need for recognition and esteem. Therefore, from the above reasons, the relationship between narcissism and features such as perceived ease of use, perceived competence, and competitive pressures should be assessed to understand the outcomes of GAI adoption. Apart from the TOE framework, narcissism is used as an essential moderator in the existing context.

Hence, the research seek to answer the following research questions:

RQ1. To what extent does the TOE framework influence the intention to adopt Generative AI?

RQ2. Is the relationship between intention to adopt GAI and competitive advantage and firm performance moderated by narcissism (high vs low)?

This research provides several important insights into the existing body of literature regarding technology adoption and organisational behaviour in the context of the hospitality industry. To begin with, it improves understanding of Generative AI (GAI) adoption by applying the Technology-Organization-Environment (TOE) framework in conjunction with the individual-level trait of narcissism, which has not been given sufficient attention, thereby providing a new multi-level lens. The study sheds important light on the interplay of a firm's competitive advantage and performance with GAI adoption vis-à-vis organisational readiness, technology capability, and environmental forces as they combine with certain traits of the decision-makers' personality. Second, the study puts forth the concept of narcissism as an important moderating factor, empirically validating its role in strategic decisions around innovation and digital advancement. Finally, the use of longitudinal design allows the research to capture shifts in the dynamics over time, thus contributing depth to theory while helping managers in planning how to optimally employ GAI to boost operational efficiency, enhance customer interactions, and achieve organisational goals.

This study contains several major chapters. Section 2 describes the nature of the problem and provides the background of the issue which is the use of 'Generative AI' in hospitality along with the research questions. Section 3 summarises the literature on the TOE framework, generative AI, its adoption, firm performance, and narcissism to construct the theory. Section 4 describes the methodology in detail, including the overall framework of the study, the longitudinal methodology for data collection, and the analysis techniques that were applied. Section 5 describes the analysis performed on the data and the results, specifically the relationships between the TOE variables, narcissism, and the consequences of GAI adoption. Finally, section 6 examines the outcomes in relation to the literature and discusses the implications of the research both on theory and practice, gives conclusions based on the limitations identified, and outlines proposed work for subsequent studies.

2. Background literature

2.1. Generative AI in hotels & resorts

Generative Artificial Intelligence (GAI) can potentially significantly impact the hospitality sector. (Dogru et al., 2025). It is a technology expected to innovate processes (Wong et al., 2023a; Koo et al., 2023). 'Generative AI' as a technology, along with supporting machine learning algorithms and data analytics, enables organisations to enhance many of their operations, including guest experience and resource allocation (Dwivedi et al., 2024; Ali et al., 2023). The scope of application of Generative AI also encompasses several dimensions in the context of the hospitality industry (Kim et al., 2023; Wong et al., 2023a). Some of these include automated check-in systems, virtual concierge, dynamic pricing, and predictive maintenance of the building (Zhang et al., 2024). Each of these innovations contributes to greater efficiency and savings for operational processes and improvement in guest experience (Wang, 2025). They offer specific services, proactively cater to guest requirements, and integrate with each phase of the guest experience (Wong et al., 2023a). On top of that, Generative AI has also been used to fuel innovation, competition, and sustainability of hotels and resorts worldwide (Wong et al., 2023a; Kim et al., 2024).

2.2. T-O-E framework

According to research conducted by Prasad Agrawal (2024), the TOE framework helps organisations better understand the technology adoption process. It concerns the interaction of technological, organisational, and environmental determinants of the adoption process (Chatterjee et al., 2021a). For example, technological factors deal with the attributes of the particular innovation, such as its degree of difficulty, fit with existing systems, and benefits (Polisetty et al., 2024). Organisational factors provide understanding regarding the domestic aspects of firms, including their structures, assets, and attributes (wael AL-khatib, 2023). These may include leadership, emphasis on the new idea or intervention, or organisational willingness and ability to adopt new ideas (Chatterjee et al., 2021a). Environmental factors are core aspects that deal with an organisation's outer environment, including external market scenarios, legal theories, and competitive forces (Awa & Ojiabo, 2016). In examining these three dimensions, the relevance of substantial resources to implementing the TOE framework is explicated further (Collado-Agudo et al., 2023). This becomes important for businesses that wish to harness the potential benefits of technology adoption and improve their performance.

2.3. Narcissism

Narcissism is a tendency that can be encountered among individuals who are self-centred and show grandiosity, consider themselves to be deserving of something, and are always seeking praise and appreciation (Christou et al., 2020). People in their working environment always have narcissistic personalities and will place themselves and their goals before their work goals (Tan & Yang, 2021). This could make one engage in organisational politics, such as self-promotion, deceptive practices, and showing less concern for workmates (Canavan, 2017). There are some necessary aspects within the leadership where their confidence and determination are positive (Yousaf et al., 2023). However, excessive narcissism has been suggested to be detrimental to team effectiveness, decision-making as well as organisational culture (Kim et al., 2018). Self-focused leaders will create a toxic culture where structures such as power dividers, favouritism, and blame will be the norms, leading to a decline in workforce satisfaction and effectiveness (Yousaf et al., 2023). Moreover, narcissism impedes managers from effectively dealing with other management aspects, such as fostering inventiveness within teams, working well with partners, and other factors related to the organisation's longevity (Woosnam et al., 2024).

The research examines the impact of narcissism on emerging technology preferences such as generative AI. Organisational inertia and company size are fundamental, but narcissism is more discomforting. Individuals with a high level of narcissism also tend to be risk-taking, inventive, and goal-oriented, as categorised by Dwivedi et al. (2024). These traits create a possibility for such individuals to be more receptive to changes such as GAI, which can eventually lead to a reduction in the resistance to change within the organisation (Wang, 2025). The link between narcissism and a high desire for recognition supports GAI's possible competitive advantage for hotels. Thus, by including narcissism as a moderator in the TOE paradigm, I may better understand how personality traits interact with disruptive technologies like GAI.

3. Development of hypotheses

3.1. *Perceived ease of use*

Perceived ease of use captures how individuals perceive a particular technology as effortless and convenient (Ozturk et al., 2016). Suppose decision-makers view GAI systems as user-friendly, intuitive, and easy to integrate into existing processes. In that case, they are more likely to develop positive attitudes toward GAI adoption (Kar et al., 2023). Perceived ease of use helps to overcome barriers associated with GAI (Ivanov et al., 2024). This, in turn, increases managers' intention to adopt GAI (Dwivedi et al., 2024). Furthermore, companies that perceive GAI as easy to use are more likely to invest resources and prioritise implementing GAI initiatives (Gupta & Yang, 2024). Gupta et al. (2024) note that this increases the organisation's readiness and leads to the desired implementation and use of GAI technologies. This leads us to put forward the following:

H1: Perceived ease of use is positively related to the Intention to adopt GAI

3.2. *Organisational competency*

In addition to the strategies a firm wishes to use, organisational competency introduces the collective understanding management of a firm, skills as well as capabilities towards maximising resource utilisation and opportunity exploitation (Yang et al., 2024). Companies with solid organisational competency can comprehend the pros and cons of GAI-sensitive technologies (Dogru et al., 2025). They are also more competent in managing the appropriate use of these technologies (Dwivedi et al., 2024). There is commensurate potential for understanding organisational control with the adoption of GAI (Dogru et al., 2025) to pursue the strategic objectives of the organisation, to be more competitive, and to improve performance (Wang, 2025).

A firm's competency is vital in shaping the decisions of hotels and resorts to adopt generative AI (Law et al., 2023). Competency is the strategic ability to see and pursue a new opportunity (Kim et al., 2023). Large hotel chains understand how valuable GAI will be to their functioning and want to adopt it to bolster their advanced AI systems (Kim et al., 2023), thus improving the customer experience (Law et al., 2023). Therefore, I incorporate the idea that a positive connection exists between organisational competency and the will to adopt Generative AI. Therefore, I hypothesise:

H2: Organizational competency is positively related to the Intention to adopt GAI

3.3. *Competitor pressure*

Competitor pressure, derived from the concern of rival organisations and market forces, provides strong reasons for organisations to adopt new technologies (Merhi & Harfouche, 2024). Numerous researchers have explored the links between perceived competitive pressure and firms' inclination to adopt technology (Tyler et al., 2020). As a result, competitor pressure will positively affect business organisations' intentions to use Generative AI (Prasad Agrawal, 2024). When one of the hotels starts

to use modern technologies, it sets a new standard, making it difficult for other hotels to ignore such technologies (Buhalis et al., 2024; Dwivedi et al., 2024; Nguyen et al., 2022). This collective rivalry forms an environmental construct within the Technology-Organization-Environment (TOE) framework (Van Huy et al., 2024). Consumers' market dynamics push Stare Jeans in GAI to retain competitiveness (Buhalis et al., 2024). Hence, I propose:

H3: Competitor pressure is positively related to the Intention to adopt GAI

3.4. Intention to adopt generative AI

Prior studies have revealed the fact that the willingness of organisations to adopt new technologies such as GAI is predicated on the ability to be competitive (Kar et al., 2023). Focusing on GAI allows organisations to be more efficient and effective in service delivery and offer customers better quality goods and services (Kanbach et al., 2024). In addition, GAI allows companies to analyse data and use them to predict trends, adjust offerings, and react to changing customers' demands quickly (Kar et al., 2023). Hence, it can be argued that organisations with aggressive GAI orientation will more likely gain and sustain a competitive advantage as innovation and market responsiveness leaders (Dwivedi et al., 2024). Thus, I hypothesise:

H4: Intention to adopt GAI is positively related to competitive advantage

According to previous research, the desire of the companies to acquire GAI reveals their proactive attitude, especially in adopting new and rapid technologies to enhance the organisation's capabilities and achieve its strategic objectives (Dwivedi et al., 2024). It can be added that GAI deployment can be beneficial to a company's performance (Kanbach et al., 2024), making better use of its resources, cutting down on expenses, growing its intake, and improving the level of contentment among clients (Hollebeek et al., 2024). Also, implementing GAI is helpful because it leads to competitive gains, which contribute toward improving financial performance and market scope (Kar et al., 2023). Thus, I hypothesise:

H5: Intention to adopt GAI is positively related to firm performance

3.5. Competitive advantage and firm performance

Competitive advantage is such a factor that supports a company's performance above its competitors. Chadwick & Flinchbaugh (2021) note that such advantages facilitate superior operational and profitability performances. Businesses that enjoy competitive strategies that the present market cannot surpass stand the chance of fierce market rivalry and earn more returns and profits (López-Gamero et al., 2009). Competitive advantages also help tackle oppression from external and circumstantial change within the industry (Yong Kim & Oh, 2004). Therefore, more competitive advantages are expected, such as better-paying policies, growing revenue, and more market share (Kumar et al., 2011). Hence, I propose:

H6: Competitive advantage is positively related to firm performance

3.6. Narcissism as a moderator

Narcissism, characterised by a self-absorbed and high need for adoration and validation (Pizam et al., 2022), could impact the leader's attitudes and motivations toward the efforts of GAI adoption. It is important to investigate how competitive advantage is leveraged to achieve firm performance in the presence of narcissistic leaders (Shirokova et al., 2024). Such leaders are more prone to acquiring GAI for prestige enhancement rather than in pursuit of winning in the market only (Budhwar et al., 2023). Moreover, those higher up the hierarchy and narcissistic may act in a manner that undermines the

efficacious adoption of GAI (Liang et al., 2024). Such behaviours include the abuse of catchphrases and catchword plans used to achieve short-term results instead of pursuing a long-term plan, especially those made by superiors (Liang et al., 2024). High levels of narcissism may dilute the positive relationship between the intention to adopt GAI and competitive advantage and firm performance (Goli & Khan, 2022). Thus, I formulate the following hypothesis:

- H7a: Narcissism will moderate the association between the intention to adopt GAI and competitive advantage
- H7b: Narcissism will moderate the association between the intention to adopt GAI and firm performance
- H7c: Narcissism will moderate the association between competitive advantage and firm performance

Our research model is shown in Figure 1.

4. Methodology

Utilising a longitudinal research design allowed us to understand better how adopting generative AI influences competitive advantage and firm performance. This study employs a longitudinal research design which allows tracking of generative AI's (GAI) impact on competitive advantage and firm performance over time. Longitudinal research is more suited for this study as it captures the technological and organisational adoption dynamics along with changing perceptions in management as opposed to snapshot analyses from cross-sectional studies. GAI, an emerging technology, is expected to have gradual impacts on organisations as they adapt, develop new competencies, and strategically evolve over time. Collecting data during two separate time intervals—prior to and following a major adoption milestone—enables the study to demonstrate progress, identify causal links, and evaluate enduring effects on performance outcomes. These insights help improve accuracy while deepening understanding on the impacts of GAI in the hospitality industry. The initial period was from November 2022 to February 2023, and the subsequent was from December 2023 to February 2024. The participants were managers (senior and middle-level employees) of hotels and resorts in India.

Participants were selected based on specific screening criteria.

- a) Are you a manager (senior, middle, or junior level) in a hotel or resort?
- b) Are you aware of Generative AI in the hospitality industry?
- c) Would you be willing to share your responses later?

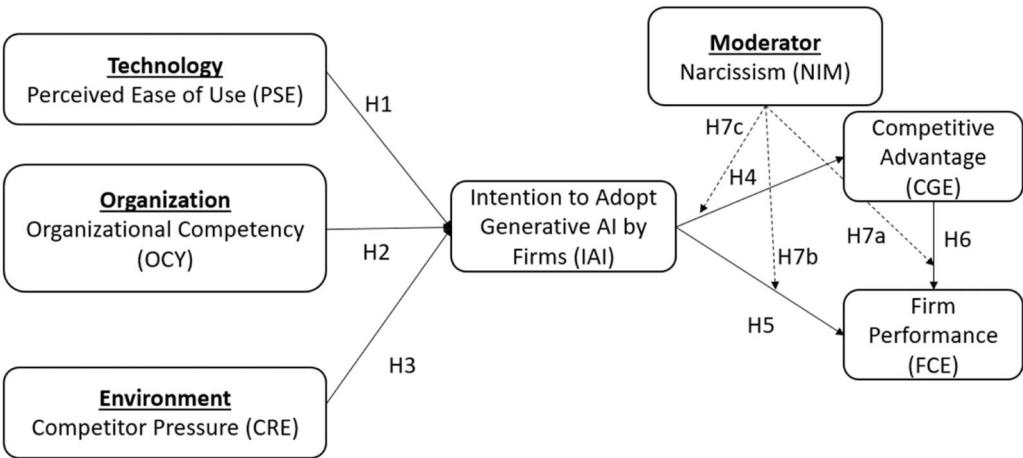


Figure 1. Hypothesised research model.

- d) Could you please share the contact information (email/WhatsApp) of your peers or relatives currently working as managers in hotels or resorts in India? (Optional)

An online survey was sent to a diverse pool of 1,487 participants from various cities across India. The survey was distributed via multiple channels such as email, LinkedIn, Facebook Pages, and WhatsApp. In the initial phase, there were 437 responses, and the second wave yielded 408 participants.

The measures were adapted from existing literature (see Table 2). The items were evaluated using a five-point scale, ranging from 1 ('strongly disagree') to 5 ('strongly agree').

Most hotels and resorts included in the study were under 10 years old. The sample was dominated by male participants (55%). A significant portion are diploma holders (55%) and junior managers (55%)% (Table 1).

5. Analysis

I used Structural Equation Modeling (SEM) with Smart PLS for this study. Using a three-stage approach, the study analysed the model's performance in different firm age contexts, specifically less than or equal to 10 years and more than 10 years. This allows for a comparison of the estimated path coefficients. Our analysis covers the evaluation of measurement models, structural models, and multi-group analysis. Before conducting the main analysis, I performed preliminary checks to ensure the reliability of the results. These checks included assessing normality and common method bias.

Before conducting hypothesis testing, it was crucial to evaluate the existence of multicollinearity. Based on the criteria established by Petter et al. (2007), the study assessed all VIF values to determine the presence of severe multicollinearity. A VIF of ≤ 3.3 (in both the time frame, t1 & t2) indicates an absence of such multicollinearity. With all VIF values below the threshold set by Petter et al. (2007), there were no significant multicollinearity issues, allowing the study to move forward with hypothesis testing confidently. Table 2 describes the factor loadings for sources, factor loadings, and constructs. Annexure 1 provides the constructs' items.

It is important to establish a measurement model before analysing the structural model. It involves verifying the convergent and discriminant validity. Convergent validity is achieved when the loading and composite reliability (CR) exceed 0.7 and the average variance extracted (AVE) is

Table 2. Constructs, sources, and factor loadings.

Constructs	Item Nos.	Sources	FL (t1)	FL (t2)
Perceived Ease of Use	PSE1	Chatterjee et al. (2021a), Erskine et al., (2019)	0.948	0.945
	PSE2		0.941	0.94
	PSE3		0.910	0.943
Organizational Competency	OCY1	Chatterjee et al. (2021a)	0.941	0.959
	OCY2		0.919	0.95
	OCY3		0.932	0.957
Competitor Pressure	CRE1	Manning et al. (2022), wael AL-khatib, (2023)	0.95	0.913
	CRE2		0.937	0.926
	CRE3		0.919	0.925
	CRE4		0.873	0.915
Intention to Adopt Generative AI by Firms	IAI1	wael AL-khatib, (2023), Wang & Zhang, (2023)	0.913	0.952
	IAI2		0.854	0.926
	IAI3		0.883	0.956
Narcissism	NIM1	Ames et al. (2006), Tandon et al. (2023)	0.923	0.954
	NIM2		0.915	0.946
	NIM3		0.864	0.942
Competitive Advantage	CGE1	Chatterjee et al. (2021a)	0.893	0.956
	CGE2		0.912	0.896
	CGE3		0.927	0.954
Firm Performance	FCE1	Chatterjee et al. (2021a)	0.923	0.947
	FCE2		0.904	0.917
	FCE3		0.893	0.95

Table 1. Demographic details.

Demographic Measures	Category	Time 1 (n = 437)		Time 2 (n = 408)	
		Frequency	Percentage	Frequency	Percentage
Organization Age	Less than or equal to 10 Years	256	58.58%	245	60.05%
	More than 10 years	181	41.42%	163	39.95%
Gender	Female	194	44.39%	188	46.08%
	Male	243	55.61%	220	53.92%
Education	Diploma	240	54.92%	240	58.82%
	Graduate	159	36.38%	159	38.97%
	Post Graduate	38	8.70%	38	9.31%
Profile of employees	Senior Manager	32	7.32%	31	7.60%
	Middle-level Manager	165	37.76%	158	38.73%
	Junior Manager	240	54.92%	219	53.68%

above 0.5. [Table 3](#) displays the findings of the convergent validity test. The CR values are above 0.7, and the AVEs are above 0.5. This confirms the presence of convergent validity for both groups.

Ensuring discriminant validity is essential to establish that a construct differs from other constructs based on empirical evidence. The discriminant validity is considered valid when the hetro-trait-monotrait (HTMT) ratio values are below 0.9. [Table 4](#) displays the findings of the HTMT analysis.

Throughout the overall sample, the VIF values were discovered to be below the recommended threshold of 3.3 in both time frames, t1 and t2 (Petter et al., 2007), suggesting no multicollinearity concerns. Afterward, the proposed hypotheses were examined, and the results, displayed in [Table 5](#), demonstrate support for all hypotheses except two in the complete sample for t1. The results from each category (less than or equal to 10 and more than 10 years) closely correspond with those of the overall sample. Interestingly, in t2, all the hypotheses are significant except one in the overall sample. After conducting hypothesis testing, the model's ability to explain the data was evaluated. The R-squared (R^2) values for the endogenous variables are 18%, 32%, and 32.5% for CGE, FCE, and IAI for t1 and 19%, 20.5%, 33.3%, respectively, in t2, indicating different levels of explanatory strength. The predictive relevance of the study was assessed using Q-squared (Q^2) values. These values are 0.199, 0.326, and 0.313 for the endogenous constructs, CGE, FCE, and IAI for t1, and 0.196, 0.167, and 0.317, indicating varying degrees of predictive relevance across the samples. The effect size (f^2) quantifies the magnitude of the impact of the independent variables on the dependent variable. According to Cohen (1992), effect sizes of 0.02 are considered small, 0.15 are considered medium, and 0.35 are considered large. Based on [Table 5](#), the f^2 values have been provided. Both groups' age of firm levels was moderately influenced by the relationship between intention to adopt GAI and competitive advantage in t1. Regarding the relationship between firms' intention to adopt GAI and competitive advantage (in t2), the study discovered that it had a small effect size. The model fit values are also provided in [Table 5](#), indicating a good model fit in both time frames.

Before performing a Multi-Group Analysis (MGA) type of analysis, it is recommended that measurement invariance is established, which will help select the best type of MGA to perform. Measurement Invariance of Composite (MICOM) suggested by Henseler et al. (2016) in Smart PLS was intended to take into the peculiarities of Smart PLS, hence distinguishing it from the covariance-based SEM approaches. The MICOM technique comprises three major components: configural invariance, assessment of compositional invariance, and equal means and variances. Groups may be compared at the first and second levels or steps, for they are said to have achieved partial measurement invariance. The group may also be compared to the group once all three steps are completed since this is called complete measurement invariance. I analysed the indicators, data treatment, and algorithm settings across two groups to ensure that they were homogeneous so that no bias would arise between them. The study successfully passed this initial step of configural invariance, thanks to the diligent and consistent recording and treatment of data. A MICOM analysis was performed for steps two and three, as outlined in Annexure 2. Based on the results, it was found that there was

Table 3. AVE, CR & reliability values.

Time 1 (t1)				
Cronbach's alpha		Composite reliability (rho_a)	Composite reliability (rho_c)	Average variance extracted (AVE)
CGE	0.897	0.899	0.936	0.829
CRE	0.939	0.944	0.957	0.847
FCE	0.892	0.902	0.933	0.822
IAI	0.861	0.885	0.915	0.781
NIM	0.884	0.89	0.928	0.812
OCY	0.923	0.924	0.951	0.866
PSE	0.926	0.926	0.953	0.871
Time 2 (t2)				
Cronbach's alpha		Composite reliability (rho_a)	Composite reliability (rho_c)	Average variance extracted (AVE)
CGE	0.928	0.933	0.955	0.875
CRE	0.939	0.941	0.956	0.846
FCE	0.932	0.935	0.956	0.88
IAI	0.94	0.941	0.961	0.893
NIM	0.943	0.946	0.963	0.898
OCY	0.952	0.954	0.969	0.913
PSE	0.937	0.939	0.96	0.889

Table 4. HTMT results.

	Time 1 (t1)						
	CGE	CRE	FCE	IAI	NIM	OCY	PSE
CGE							
CRE	0.506						
FCE	0.369	0.567					
IAI	0.457	0.52	0.464				
NIM	0.221	0.396	0.467	0.34			
OCY	0.463	0.583	0.606	0.553	0.453		
PSE	0.555	0.619	0.585	0.514	0.393	0.615	
Time 2 (t2)							
	CGE	CRE	FCE	IAI	NIM	OCY	PSE
CGE							
CRE	0.292						
FCE	0.317	0.298					
IAI	0.365	0.516	0.382				
NIM	0.333	0.289	0.242	0.364			
OCY	0.327	0.434	0.405	0.481	0.359		
PSE	0.353	0.384	0.334	0.357	0.238	0.29	

compositional invariance. This was supported by permutation *P*-values greater than 0.05, indicating that the original correlation for each variable did not differ significantly from 1. In the third step of the study, full measurement invariance was also achieved. This was evident from the permutation *P*-value for the confirmation construct, which was more than 0.05. This suggests that there were no significant differences between the age of the firm below or equal to 10 years and more than 10 years. As a result, the study achieved full invariance, which allowed for a comparison between the entire group.

No significant distinctions were found between the results for both groups when I saw the Multiple Group Analysis (MGA) results. Utilising PLS-MGA, disparities were examined by applying the MGA test to datasets of both categories. The differences in path coefficients between the two datasets are presented in Table 6. Out of the eleven hypotheses for MGA, all were found to be unsupported in both time frames. Thus, the findings did not support H1, H2, H3, H4, H5, H6, H7, H8, H9a, H9b and H9c. The differences in path coefficients between the two datasets are outlined in Table 6.

6. Discussion

AI-powered chatbots that handle guest inquiries and reservations, virtual concierges that offer personalised recommendations during a guest's stay, and dynamic pricing systems that optimise hotel rates based on market demand are examples of GAI in hospitality. Generative AI improves visitor experiences, operations, and competitiveness, as shown in these cases. For numerous reasons, perceived simplicity of use positively affects hotels' and resorts' propensity to implement Generative AI (H1). Generational AI's focus on reducing processes and improving efficiency makes simplicity of use a key adoption factor, matching stakeholders' desire for user-friendly technologies (Gupta et al., 2024). Many factors affect the positive relationship between organisational competency and Generative AI intention (H2). Organisational competency, capability demonstration, and resources affect GAI readiness (Camisón et al., 2016). Hotels and resorts recognise the value of technology for competitive guest experiences and organisational capability in GAI adoption. Many factors (H3) explain the link between competitive pressure and Generative AI (GAI) adoption. Hotels and resorts must constantly satisfy guests (Lee et al., 2003). Generative Artificial Intelligence (GAI) adoption by competitors may inspire other companies to follow suit. As GAI's benefits and uses are recognised and adopted in the industry, competitors may increase pressure, enhancing its positive link with hotels and resorts' GAI use.

Table 5. Hypothesis results.

Time 1 (t1) (Complete)								
Hypothesis Relationship	Beta	T value	p Value	2.50%	97.50%	f square	VIF	Decision
H6: CGE -> FCE	0.151	3.107	0.002	0.055	0.244	0.027	1.226	Supported
H3: CRE -> IAI	0.222	3.504	0	0.097	0.348	0.044	1.679	Supported
H4: IAI -> CGE	0.376	8.426	0	0.282	0.459	0.155	1.112	Supported
H5: IAI -> FCE	0.249	5.168	0	0.151	0.341	0.071	1.289	Supported
H2: OCY -> IAI	0.272	4.668	0	0.154	0.385	0.067	1.647	Supported
H1: PSE -> IAI	0.182	2.993	0.003	0.064	0.299	0.028	1.744	Supported
H7a: NIM x CGE -> FCE	0.034	0.728	0.467	−0.054	0.127	0.002	1.229	Not Supported
H7c: NIM x IAI -> CGE	−0.08	1.781	0.075	−0.165	0.008	0.008	1.015	Not Supported
H7b: NIM x IAI -> FCE	−0.178	3.477	0.001	−0.277	−0.076	0.038	1.224	Supported
	R-square	Q ² predict	SRMR= 0.041					
CGE	0.18	0.199	NFI = 0.885					
FCE	0.32	0.326						
IAI	0.325	0.313						
Time 2 (t2) (Complete)								
Hypothesis Relationship	Beta	T value	p Value	2.50%	97.50%	f square	VIF	Decision
H6: CGE -> FCE	0.182	3.409	0.001	0.075	0.283	0.033	1.27	Supported
H3: CRE -> IAI	0.317	5.451	0	0.2	0.43	0.115	1.309	Supported
H4: IAI -> CGE	0.257	5.714	0	0.167	0.344	0.072	1.135	Supported
H5: IAI -> FCE	0.235	4.445	0	0.133	0.337	0.054	1.277	Supported
H2: OCY -> IAI	0.288	5.487	0	0.181	0.387	0.101	1.231	Supported
H1: PSE -> IAI	0.141	2.914	0.004	0.045	0.236	0.025	1.177	Supported
H7a: NIM x CGE -> FCE	0.182	4.064	0	0.096	0.271	0.037	1.012	Supported
H7c: NIM x IAI -> CGE	0.092	1.783	0.075	−0.007	0.195	0.009	1.126	Not Supported
H7b: NIM x IAI -> FCE	0.152	3.324	0.001	0.06	0.241	0.025	1.132	Supported
	R-square	Q ² predict	SRMR = 0.032					
CGE	0.19	0.196	NFI = 0.908					
FCE	0.205	0.167						
IAI	0.333	0.317						

Several reasons explain the positive correlation between enterprises' ambition to use Generative AI (GAI) and competitive advantage (H4). First, GAI adoption shows hotels and resorts' forward-thinking use of cutting-edge technologies to improve operational efficiency, service quality, and guest experiences. By adopting GAI, hotels and resorts can improve visitor relationships, revenue management, and competitiveness. The positive relationship between intention to adopt GAI and firm performance (H5) shows managers' dedication to using innovative technology to improve operational efficiency, guest experiences, and performance measures. This relationship's longevity shows stakeholders' knowledge of technology adoption's benefits, underlining the need to continue adopting new technologies to be competitive and succeed (Pencarelli, 2020).

Several factors drive the favourable association between competitive advantage and firm performance (H6). Firstly, hotels and resorts differentiate themselves through unique strengths, exceptional service, and efficient operations, attracting and retaining customers, leading to increased revenue and profitability. Through strategies like unique location, strong branding, and outstanding customer service, hotels and resorts set themselves apart from competitors, enabling them to charge higher prices and enhance customer satisfaction, thereby improving financial performance. Using Generative AI (GAI) technologies [intention formation, adoption, use], Managers may have different attitudes toward implicating GAI (H7a). By t1, there is no moderation effect, as illustrated, suggesting that GAI adoption intentions and competitive advantage relationship may be affected by other factors not discussed above. However, at time t2, the moderating effect may indicate a shift in leadership perspectives or external influences, such as heightened competition or technological advancements, leading narcissistic leaders to play a more significant role in shaping strategic decisions regarding GAI adoption and competitive positioning.

However, regarding the intention to embrace Generative AI (GAI) and the company performance (H7b), they mean that the lower-level factors still impact the organisation. Leaders' narcissism

Table 6. Multi-group analysis.

Time 1 (t1)			
Relationship	PLS MGA		Decision
	Difference (Fage_H10 – Fage_L10)	P value (2-tailed)	
CGE -> FCE	−0.064	0.515	Unsupported
CRE -> IAI	−0.085	0.504	Unsupported
IAI -> CGE	−0.052	0.564	Unsupported
IAI -> FCE	−0.019	0.85	Unsupported
OCY -> IAI	0.042	0.722	Unsupported
PSE -> IAI	0.101	0.408	Unsupported
NIM x CGE -> FCE	−0.144	0.117	Unsupported
NIM x IAI -> CGE	0.02	0.831	Unsupported
NIM x IAI -> FCE	0.109	0.276	Unsupported
Time 2 (t2)			
Relationship	PLS MGA		Decision
	Difference (Fage_H10 – Fage_L10)	P value (2-tailed)	
CGE -> FCE	−0.087	0.421	Unsupported
CRE -> IAI	0.142	0.227	Unsupported
IAI -> CGE	−0.189	0.051	Unsupported
IAI -> FCE	0.029	0.796	Unsupported
OCY -> IAI	−0.125	0.233	Unsupported
PSE -> IAI	0.018	0.865	Unsupported
NIM x CGE -> FCE	0.038	0.687	Unsupported
NIM x IAI -> CGE	−0.059	0.588	Unsupported
NIM x IAI -> FCE	−0.087	0.36	Unsupported

induces GAI to enhance their image among the masses. Such narcissistic leaders may have a greater drive to achieve their self-interests and spend more resources on GAI technology. This moderating effect explains the nexus between narcissism in leadership and the formulation of strategy. However, in the case of narcissism, personal competitiveness does not influence firm performance within the hotel and resort field (H7c). Thus, it is possible and sensible to consider that other factors may be more critical. Decision-making and allocation of strategic focus family and Institute internal and external resources for advantage and competitive success for the company. The exclusion of narcissistic moderation of the two-time points cites the nature of many facets of the organisation and their contribution towards competition and performance in the tourism and hospitality sector.

According to the results of this research, the age of a firm does not change specific interrelations among other variables about the hotels' and resorts' Generative AI adoption. Likewise, the study found no significant difference in the relationships between the intention to adopt Generative AI, competitive edge, and firm performance among firms of various ages. In other words, the potential of using Generative AI to enhance advantages and improve result strategies from the customers' perspective brings no difference across the industry's several age categories of resorts and hotels. Therefore, the present research emphasises the importance of extending beyond the firm age to determine the overall progression in the tourism and hospitality industry regarding technology adoption.

Drawing from these results, this research expands upon existing literature by explaining how the alignment of dynamic technological capabilities with organisational readiness can be strategically utilised to create and sustain a competitive advantage in the hospitality industry. Resorts and hotels that adopt generative AI (GAI) not only enhance their real-time decision-making but also foster brand loyalty through the provision of deeply personalised, immersive guest interactions. The adoption of GAI alongside customer engagement strategies increases customer interaction, while automated backend functions like housekeeping, inventory, and customer service, claim substantial cost savings. These efficiencies provide businesses with the opportunity to recalibrate their focus toward innovation and guest satisfaction, garnering further benefits from tech-enabled competitive growth. Additionally, GAI-enabled systems can help mitigate understaffing and training lags in notoriously high-contact services roles, further amplifying employee productivity and satisfaction.

It is particularly noteworthy how the investigation of narcissism as a moderating variable provides a distinct psychological perspective to understand strategic decision-making concerning the adoption of GAI technologies. Transformative innovations like GAI may garner early adoption and advocacy from narcissistic leaders, as they tend to seek out high-profile GAI for purposes of acquiring status and recognition. However, the results caution that while image-seeking behaviour stemming from narcissism may accelerate adoption, the benefits to structural firm performance are not always guaranteed. Such leaders may be more focused on the superficial integration of resources rather than the sustainable implementation required, causing a technologically overstretched and organisationally unrested environment. The lack of meaningful moderation in some of the relationships analysed, as well as in different age groups within firms, suggests that although personality attributes such as narcissism moderate technology decision-making, more systemic factors—like digital readiness, unified leadership, and holistic departmental engagement—are more critical for enduring performance and industry competitiveness in a highly dynamic context like hospitality.

7. Theoretical implications

First of all, with the help of the T-O-E framework, this article examines the adoption of Generative AI (Agrawal, 2024). It studies the intricate relationship among the technological aspects (Polisetty et al., 2024), the structural factors within the organisation (Chatterjee et al., 2021a), and the context of use (Awa & Ojiabo, 2016) through the lens of an adoption process. The longitudinal method helps researchers grasp the changing nature of the competitive edge in tourism and hospitality. This method helps provide a clearer picture of GAI's roles in strategic positioning and market performance.

Second, the paper has value for the hospitality industry by highlighting the competitive usage of Generative AI, particularly strategic fit and change (Merhi & Harfouche, 2024). This paper also demonstrates how organisational aspects play out in GAI use in different contexts. It examines how perceived usefulness, organisational resources, and external competition pressure influence this process (Kar et al., 2023). This study also focuses on the intra-organisational factors in the contexts of innovation projects and their results, enriching technology adoption scholarship. Using the TOE model, this construct enables one to examine different technological developments, organisational resources, and environmental aspects that determine the competitive advantage the hotels and resorts achieve as they adopt GAI technology.

Third, this research contributes to the scholarly conversation by incorporating psychological traits at the individual level—narcissism in particular—into the structurally oriented TOE framework, providing a blended sociological and psychological perspective on technology adoption in the hospitality sector. By looking into the effects of narcissistic leadership on the relationship between the adoption of GAI and the performance of a firm, the study integrates personality-driven innovative motivations with strategic decision-making. In demonstrating these relationships, the study not only reveals the strategic and operational significance of GAI, but also illuminates the ways in which leadership traits can enhance or distort its adoption. Thus, the study contributes to the body of literature on the integration of technology and leadership by showing that the successful adoption of GAI and the impact on performance in a firm relies not only on external and internal conditions of appropriateness, but the cognitive structures and motivating factors of the firm's decision-makers. Having this perspective sheds light on the wide-reaching impacts of digital transformation on highly serviced sectors like tourism and hospitality.

8. Practical implications

Managers and owners of organisations in the hospitality industry must appreciate the value of embracing Generative AI (GAI). First, with the Zamindari and the advent of GAI technologies, there is an opportunity to reduce operational efficiencies, improve customer relations, and add

more advantages over other players. Second, it is necessary to achieve this in practice by creating an environment that promotes innovative thinking and continuous education, which is necessary to deploy GAI effectively. In order to maximise the returns on GAI tools adoption, managers and owners need to ensure that every employee gets enough training and development. Third, managers and owners must ensure that organisational readiness and change management are factors considered when implementing GAI technologies.

Communication is one of the most critical areas of management research and practice; defining clear roles, responsibilities, channels, and processes, cooperation between the departments, including collaboration between non-adjacent departments, and potential organisational change resistance from the employees is essential. Fourth, it is crucial for managers and owners that competitive intelligence is sought and explored in terms of customers, their orientations, and technology developments to seize and make favourable circumstances. Trusted consultative technology providers and industry networks may provide support and further guidance on the strategy development and adoption phases. Soon, through the effective use and integration of the GAI tools, managers and owners will improve their operational effectiveness, improve customer satisfaction, and prepare their hotels and resorts for successful operations in the highly competitive environment and the calendar market.

Moreover, alignment of strategic objectives as well as evaluation from the market requires the constant active attention from the managers and the owners regarding the outcomes of GAI execution. Performance metrics such as customer satisfaction, the ratio of bookings to inquiries, operational savings, and employee productivity will measure GAI's performance to the company's goals. In addition, building feedback and agility can enhance process ownership among employees, thus encouraging AI process refinement which leads to innovation and improvement. Also, managers must monitor the growing concerns of trust issues on the brand especially in the discipline of AI ethics, data privacy, and law because these domains are critical to customer loyalty and corporate image. Addressing these issues, the generals in hospitality can use marketing strategies to ensure GAI investments are not only short gain but also sustain adaptability and competitiveness in the healthcare industry's aging digital environment.

9. Limitations and future research

Though the findings of this research indeed point towards generating valuable insights for practice and theory alike, the applicability of these results is affected by certain assumptions and other aspects unique to the study. Therefore, the ability to transfer these conclusions to other hospitality environments or industries will be limited. Also, since most GAI technologies are complex, changing rapidly, and becoming more advanced, it is prudent to continuously investigate their effects on organisational performance, customer satisfaction, or employee relations. Subsequent studies may consider organisational culture and leadership, market conditions, how these mediate GAI adoption and implementation, and their subsequent impact on competitive advantage.

10. Conclusion

The advancing technologies of Generative AI (GAI) significantly transforms the means by which hotels and resorts compete and innovate in the context of preserving consumer digital attention and experience. From a narcissism standpoint, especially integrated with a TOE approach, this study offers a detailed understanding of GAI adoption's multi-faceted influences. The longitudinal and cross-sectional multi-group approach provides insights into the stages of organisational preparedness, technological readiness, environmental conditions, and personality traits of decision makers which greatly influence the application of GAI by the firm and its realisation on the competition and performance of the firm. The refinement of the TOE framework concerning the contribution

strengthens the literature on hospitality innovation by integrating social cognitive factors, thus enhancing the understanding of innovation adoption behaviours in hospitality business.

This research may be of practical use to hospitality managers, business owners, and government decision-makers. It was observed that organisational preparations with regard to structures, technologies, and leaders' mindsets, as well as their personality traits, are significant for the successful adoption of GAI. It was noted that high narcissism can be detrimental but, if contained, can also foster rapid and bold strides towards innovation. This highlights the dual governance needed to maintain confidence on one hand and avoid excessive exposure to needless risk on the other. Consequently, hospitality firms are urged to analyse both the human and technological dimensions when devising strategies for digital transformation. With GAI's internal and external drivers comprehensive stakeholder efforts can be channelled towards strategically sculpted interventions aimed to heighten guest and operational interface while reinforcing agile edge advantage within a transforming services environment.

Disclosure statement

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